

Joan Alcaide-Núñez

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EDUCATION

Ludwig-Maximilians-Universität (LMU) Munich

B.Sc. in Physics

Munich, Germany

Oct 2025 – Present

Core concentration in foundational and mathematical physics pathways.

German School of Barcelona

Dual High School Diploma (Germany + Spain)

Barcelona, Spain

Sept 2017 – May 2025

Academic Achievement Award with a perfect Grade Point Average of 1.0 / 1.0.

Additional Specialised Training:

- *Gravity and Black Holes* – Perimeter Institute for Theoretical Physics (GoPhysics!), 2025
- *Mathematics, Statistics, and Data Science* – Autonomous University of Barcelona (UAB), 2022
- *Algorithms and Programming (C++)* – Polytechnic University of Catalonia (UPC), 2022
- *League of Codes (C++ Masterclass)* – Harbour Space University, 2022 – 2023

RESEARCH EXPERIENCE

Theoretical Probes of Spacetime

Independent Research Project

Remote / Independent

July – Sept 2026

- Derived Modified Dispersion Relations (MDR) for high-energy photons traveling through non-standard cosmological backgrounds to probe potential Planck-scale violations.
- Applied Hamilton-Jacobi equations to analytically calculate cumulative time-of-flight delays (Δt) for gamma-ray bursts propagating under Quintessence-based FLRW metrics.
- Investigated the mathematical coupling parameter between the dark energy equation of state (w) and energy-dependent group velocities.

GRB Cosmography & Relativistic Foundations

Osservatorio Astronomico di Brera (OAB-INAF)

Merate, Italy

June – July 2025

- Constrained global cosmological parameters (Ω_m, Ω_Λ) via χ^2 minimisation profiles using the empirical $E_{\text{peak}} - E_{\text{iso}}$ relation of Gamma-Ray Bursts.
- Modeled detection synergies with mock binary neutron star populations matching Einstein Telescope specifications.
- Received foundational technical training on relativistic jet kinematics from Dr. Gabriele Ghisellini.
- *Supervisors:* Dr. Giancarlo Ghirlanda, Dr. Om Sharan Salafia · [Repository Link](#).

PI GAMMA Project

Founder & Project Lead – CAPIBARA Collaboration

International

2024 – Present

- Established the CAPIBARA Collaboration to connect science-driven international students across secondary and university levels to coordinate collaborative learning frameworks.
- Leading the [GAMMA Team](#) evaluating the technical feasibility, instrumentation parameters, and mission profiles for a student-designed gamma-ray transient observing CubeSat.

Type Ia Supernovae: Mapping Expansion History

Institute of Space Sciences (ICE-CSIC-IEEC)

Barcelona, Spain

June – July 2024

- Extracted infrared imaging data from ESO (SOFI) and cross-matched optical photometry from ZTF and ATLAS using the `sncosmo` Python suite to map luminosity distances.
- Explored non-linear expansion metrics, solving the Friedmann acceleration equations to fit cosmic density parameters via light-curve parameterisation.
- *Supervisor:* Dr. Lluís Galbany.

Cepheid Variable Stars Research

Research Fellow – Youth & Science Program

Budapest, Hungary

Aug – Dec 2023

- Reconstructed classical Hubble-Lemaître expansion constraints, deriving an empirical value of $H_0 = 70.26 \pm 7.1 \text{ km s}^{-1} \text{ Mpc}^{-1}$ using modern Cepheid observations from the Konkoly Observatory.

- Selected for a selective 3-year funded research track, marking first formal exposure to academic data review · [Technical Report](#).
- *Supervisors:* Dr. Ignasi Pérez-Ràfols, Dr. Laia Casamiquela.

German Aerospace Center (DLR)

Research Intern – Institute for Remote Sensing (DLR-IMF)

Oberpfaffenhofen, Germany

Oct 2024

- Assisted photogrammetry teams with high-dimensional hyperspectral imaging calibrations and automated Land-Sat thermal layer array analysis.

AWARDS & HONORS

Youth & Science Fellowship · *Fundació Catalunya La Pedrera*

2023 – 2026

Fully funded 3-year international research fellowship supporting consecutive competitive summer research placements.

Abiturpreis (Physics Distinction) · *German Physical Society (DPG)*

2025

National prize awarded for outstanding academic excellence and foundational focus in physics during high school.

International Astronomy and Astrophysics Competition (IAAC)

2023 – 2025

Awarded the **Silver Honour** and the **National Award for Spain** (2025), ranking in the top 7% globally out of more than 12,000 international participants.

Jugend Forscht (Youth Research Competition)

2024 – 2025

Awarded **First Prize Iberia** across regional and national tiers; received the Special Award in Scientific Photography (NRW, 2025).

TALKS & SCIENCE OUTREACH

- **SSEA Symposium (April 2026):** Presented “*The CRD Instrument and the COSMOS Program.*”
- **CosmoXarxa (March 2025):** Delivered an invited presentation on high-energy astrophysics metrics at the Barcelona Science Museum.
- **Academic Workshops (2023 – 2024):** Designed introductory astrophysics workshops for 11th graders at the German School Barcelona and organized planetary space exploration seminars for elementary tracks.
- **Museum Explainer (2022 – 2023):** Content volunteer conducting interactive public physics experiments at the CosmoCaixa Science Museum.

WORK EXPERIENCE

Ludwig-Maximilians-Universität (LMU) Munich

Student Assistant – Central Diversity Management & Research Funding Unit

Munich, Germany

Oct 2025 – Present

TECHNICAL SKILLS

Programming & Core Frameworks: Python (*NumPy*, *SciPy*, *Pandas*, *Matplotlib*, *Astropy*), **sncosmo**, C++, L^AT_EX, Markdown.

Tools & Environments: Git, GitHub, Visual Studio Code, NeoVim, Unix/Linux command-line environments.